

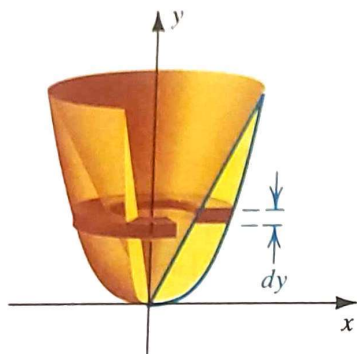
FIGURE 6.28
(iii)

Figure 6.28(ii) illustrates the volume generated by the region and the washer generated by the rectangle. We note the following:

thickness of washer: dy

outer radius: $2y^{1/3}$

inner radius: $\frac{1}{2}y$

$$\text{volume: } \pi[(2y^{1/3})^2 - (\frac{1}{2}y)^2] dy = \pi(4y^{2/3} - \frac{1}{4}y^2) dy$$

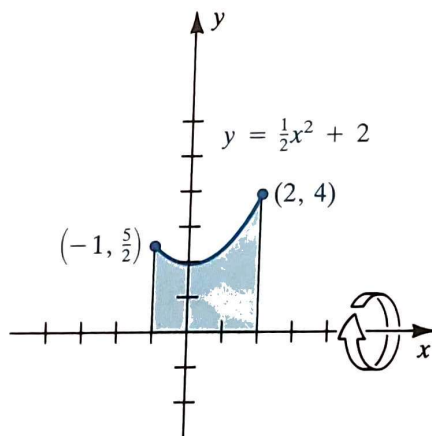
Applying the limit of sums operator $\int_0^8$ gives us the volume:

$$\begin{aligned} V &= \int_0^8 \pi(4y^{2/3} - \frac{1}{4}y^2) dy \\ &= \pi \left[\frac{12}{5}y^{5/3} - \frac{1}{12}y^3 \right]_0^8 = \frac{512}{15}\pi \approx 107.2 \end{aligned}$$

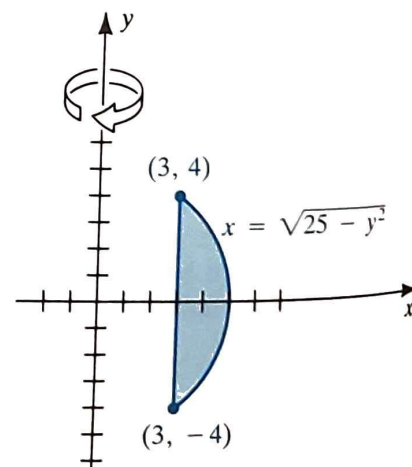
EXERCISES 6.2

Exer. 1–4: Set up an integral that can be used to find the volume of the solid obtained by revolving the shaded region about the indicated axis.

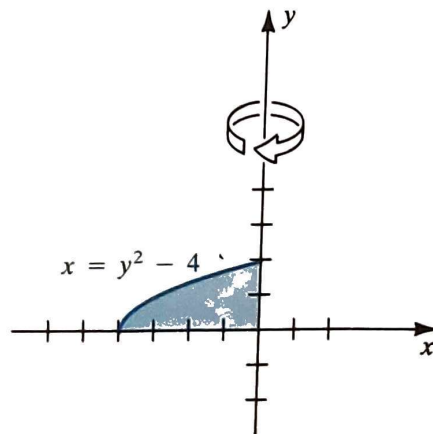
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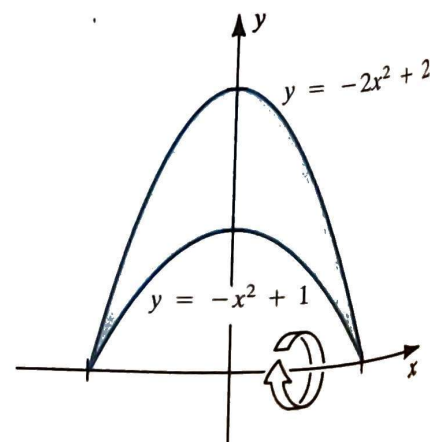
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4



Exer. 5–24: Sketch the region R bounded by the graphs of the equations, and find the volume of the solid generated if R is revolved about the indicated axis.

5 $y = 1/x$, $x = 1$, $x = 3$, $y = 0$; x -axis

6 $y = \sqrt{x}$, $x = 4$, $y = 0$; x -axis

7 $y = x^2 - 4x$, $y = 0$; x -axis

8 $y = x^3$, $x = -2$, $y = 0$; x -axis

9 $y = x^2$, $y = 2$; y -axis

10 $y = 1/x$, $y = 1$, $y = 3$, $x = 0$; y -axis

11 $x = 4y - y^2$, $x = 0$; y -axis

12 $y = x$, $y = 3$, $x = 0$; y -axis

13 $y = x^2$, $y = 4 - x^2$; x -axis

14 $x = y^3$, $x^2 + y = 0$; x -axis

15 $y = x$, $x + y = 4$, $x = 0$; x -axis

16 $y = (x - 1)^2 + 1$, $y = -(x - 1)^2 + 3$; x -axis

17 $y^2 = x$, $2y = x$; y -axis

18 $y = 2x$, $y = 4x^2$; y -axis

19 $x = y^2$, $x - y = 2$; y -axis

20 $x + y = 1$, $x - y = -1$, $x = 2$; y -axis

21 $y = \sin 2x$, $x = 0$, $x = \pi$, $y = 0$; x -axis

(Hint: Use a half-angle formula.)

22 $y = 1 + \cos 3x$, $x = 0$, $x = 2\pi$, $y = 0$; x -axis

(Hint: Use a half-angle formula.)

23 $y = \sin x$, $y = \cos x$, $x = 0$, $x = \pi/4$; x -axis

(Hint: Use a double angle formula.)

24 $y = \sec x$, $y = \sin x$, $x = 0$, $x = \pi/4$; x -axis

Exer. 25–26: Sketch the region R bounded by the graphs of the equations, and find the volume of the solid generated if R is revolved about the given line.

25 $y = x^2$, $y = 4$

(a) $y = 4$ (b) $y = 5$

(c) $x = 2$ (d) $x = 3$

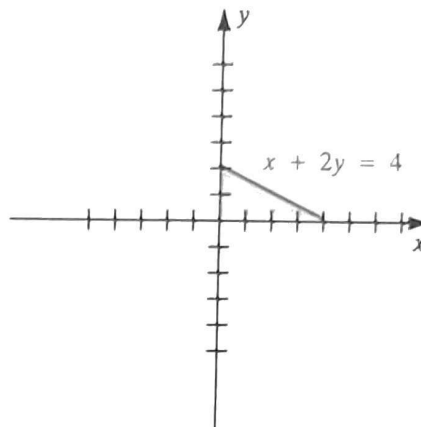
26 $y = \sqrt{x}$, $y = 0$, $x = 4$

(a) $x = 4$ (b) $x = 6$

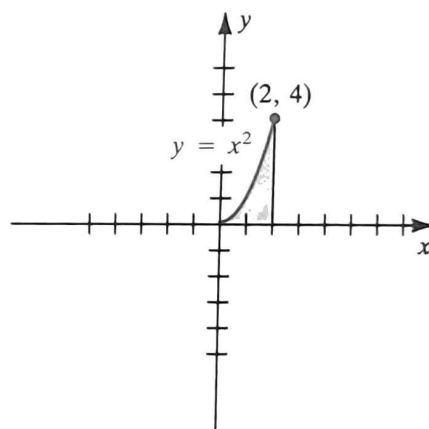
(c) $y = 2$ (d) $y = 4$

Exer. 27–28: Set up an integral that can be used to find the volume of the solid generated by revolving the shaded region about the line (a) $y = -2$, (b) $y = 5$, (c) $x = 7$, and (d) $x = -4$.

27



28



Exer. 29–34: Sketch the region R bounded by the graphs of the equations, and set up integrals that can be used to find the volume of the solid generated if R is revolved about the given line.

29 $y = x^3$, $y = 4x$; $y = 8$

30 $y = x^3$, $y = 4x$; $x = 4$

31 $x + y = 3$, $y + x^2 = 3$; $x = 2$

32 $y = 1 - x^2$, $x - y = 1$; $y = 3$

33 $x^2 + y^2 = 1$; $x = 5$

34 $y = x^{2/3}$, $y = x^2$; $y = -1$

Exer. 35–40: Use a definite integral to derive a formula for the volume of the indicated solid.

35 A right circular cylinder of altitude h and radius r

36 A cylindrical shell of altitude h , outer radius R , and inner radius r

37 A right circular cone of altitude h and base radius r

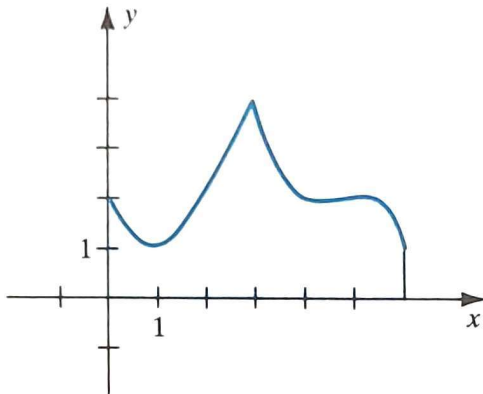
38 A sphere of radius r

39 A frustum of a right circular cone of altitude h , lower base radius R , and upper base radius r

40 A spherical segment of altitude h in a sphere of radius r

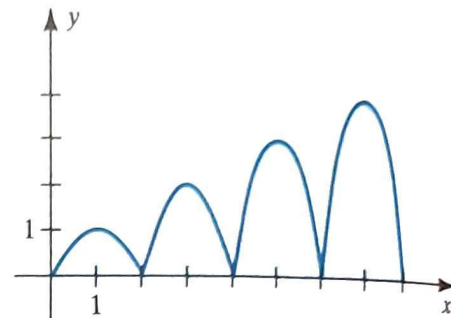
- 41 If the region shown in the figure is revolved about the x -axis, use the trapezoidal rule with $n = 6$ to approximate the volume of the resulting solid.

EXERCISE 41



- 42 If the region shown in the figure is revolved about the x -axis, use Simpson's rule with $n = 8$ to approximate the volume of the resulting solid.

EXERCISE 42



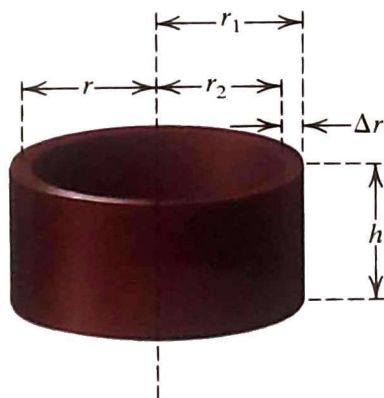
- [c] Exer. 43–44: Graph f and g on the same coordinate axes for $0 \leq x \leq \pi$. (a) Estimate the x -coordinates of the points of intersection of the graphs. (b) If the region bounded by the graphs of f and g is revolved about the x -axis, use Simpson's rule with $n = 4$ to approximate the volume of the resulting solid.

$$43 \quad f(x) = \frac{\sin x}{1+x}; \quad g(x) = 0.3$$

$$44 \quad f(x) = \sqrt[4]{|\sin x|}; \quad g(x) = 0.2x + 0.7$$

6.3 VOLUMES BY CYLINDRICAL SHELLS

FIGURE 6.29



Volume V of a
cylindrical shell (6.10)

In the preceding section we found volumes of solids of revolution by using circular disks or washers. For certain types of solids it is convenient to use hollow circular cylinders—that is, thin **cylindrical shells** of the type illustrated in Figure 6.29, where r_1 is the *outer radius*, r_2 is the *inner radius*, h is the *altitude*, and $\Delta r = r_1 - r_2$ is the *thickness* of the shell. The *average radius* of the shell is $r = \frac{1}{2}(r_1 + r_2)$. We can find the volume of the shell by subtracting the volume $\pi r_2^2 h$ of the inner cylinder from the volume $\pi r_1^2 h$ of the outer cylinder. If we do this and change the form of the resulting expression, we obtain

$$\begin{aligned} \pi r_1^2 h - \pi r_2^2 h &= \pi(r_1^2 - r_2^2)h \\ &= \pi(r_1 + r_2)(r_1 - r_2)h \\ &= 2\pi \cdot \frac{1}{2}(r_1 + r_2)h(r_1 - r_2) \\ &= 2\pi r h \Delta r, \end{aligned}$$

which gives us the following general rule.

$$V = 2\pi(\text{average radius})(\text{altitude})(\text{thickness})$$

If the R_x region in Figure 6.30(i) is revolved about the y -axis, we obtain the solid illustrated in (ii) of the figure.